

APPLIED SCIENCES AND ENGINEERING

High-speed fluidic processing circuits for dynamic control of haptic and robotic systems

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Fluidic logic circuits simplify system design for soft robotics by eliminating bulky components while enabling operation in a range of hostile environments that are incompatible with electronics but at the expense of limited computational capabilities and response times on the order of seconds. This paper presents a four-terminal fluidic transistor optimized for fast switching times, reduced component count, low unit cost, and high reproducibility to achieve complex fluidic control circuits while maintaining flow rates of liters per minute. A ring oscillator using three fluidic transistors achieves oscillation frequencies up to a kilohertz with full signal propagation, tolerating billions of cycles without failure. Fundamental processor circuits like a full adder and a 3-bit analog-to-digital converter require just seven transistors each. A decode circuit drives a high-resolution soft haptic display with refresh times below the human perception threshold for latency, and an electronics-free control circuit performs closed-loop position control of a pneumatic actuator with disturbance rejection, demonstrating the value across domains.

INTRODUCTION

Biological inspiration drove many of the early advances in the field of soft robotics (1) as researchers sought to replicate many of the ways in which nature outperformed conventional, rigid robots, from deformability and compliance for safe interaction with humans to resistance to impact (2, 3). Demonstrations ranged from soft snakes (4), manta rays (5), octopus arms (6, 7), muscle tendons (8), fish with electrolytic vascular systems (9), and multi-gate walking robots (10) to capabilities like camouflage (11) and self-healing (12). As the technology progressed, a wide range of applications emerged, including joint rehabilitation (13), ventricular assist devices (14), surgical robotics (15), and a variety of grippers (16) capable of handling even the most delicate of objects (17).

The physical tethers, both pneumatic and electrical, presented a persistent caveat to many early soft robotics demonstrations, particularly the mobile robots. In an early wave of untethered soft robots, the vehicles were simply built large enough to accommodate the necessary electronics, valves, regulators, and pressure supplies onboard. However, advances in soft sensing, pressure generation, and electronics soon made fully soft untethered devices feasible (18), such as an octopus filled with decomposing hydrogen peroxide driving a bistable fluidic oscillator to wave its arms (19). For higher energy density, various robots powered themselves through controlled explosions or combustion (20–22). Stretchable optical waveguides (23) or embedded electrical resistive sensors combined with neural networks (24) can measure large strains reliably.

Electronics-free robots could function without degradation or risk of sparking in nuclear disaster sites, underground mines, outer space, or magnetic resonance imaging machines, motivating researchers to move all the system-level components out of the electrical domain and into the fluidic and chemical domains. Highly deformable strain gauges with an embedded flow channel (25) sense changes in pneumatic resistance as the sensor strains, creating a fluidic analogy for electrical strain gauges. Even motors have been demonstrated in the pneumatic domain with an inflatable stator

driving a rotor through a peristaltic motion (26). To satisfy the operational definition of a robot (27), these devices must not only sense and act but also think with the discipline of fluidic logic showing the most promise for bridging this gap without electronics.

In the field of microfluidics (28), advancements in multilayer soft lithography more than two decades ago led to pressure-piloted “Quake-style” fluidic valves (29) and microfluidic large-scale integration (30). With this style of fluidic logic, a small number of external solenoid valves could control a larger number of fluidic outputs through demultiplexers and shift registers (31) for applications like single-cell functional analysis (32). Inclusion of a rigid SU-8 disk in the elastomer of the valve gate enabled pressure gain for cascable logic (33), and fluidic oscillators provided the basis for clock timing on the chip (34, 35), where the resistive-capacitive (RC) dynamics could be tuned by adding external capacitance to slow the oscillator (36) or adjusting the pull-down resistor to speed up the oscillator to a maximum of 50 Hz (37), albeit at the expense of reducing the output signal range. Recent work exploited the fluidic phenomenon of flow limitation to mimic a transistor’s saturation behavior, enabling a particle dispenser circuit without electronics (38). While these microfluidic logic circuits achieved notable success in academic labs, the peripheral electronics equipment required to drive them, from pressure regulators and sensors to solenoid valves and digital acquisition systems, has led to a relative lack of commercial success as some critics describe the resulting system as more of a “chip-in-a-lab” than a “lab-on-a-chip” (39). However, these early learnings in microfluidics paved the way for the field of soft robotics to push further toward fully fluidic computation in recent years.

While microfluidics logic circuits built on the micron scale typically work with flow rates ranging from nanoliters to milliliters per minute, soft robotics logic circuits require higher flow rates to enable higher forces and faster actuations, leading researchers to build fluidic valves on the millimeter or even centimeter scale (40). These early soft valves harness snap-through buckling instabilities to alternate the kinking of two distinct flow channels in a six-terminal device, enabling a range of digital logic gates (41) and nonvolatile memory (42) as well as a fully soft ring oscillator (43). The inherent complimentary logic of these valves and other CMOS-inspired

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fluidic valves (44) eliminated some of the challenges associated with resistor-transistor logic typically used in microfluidic logic such as steady-state power consumption and the reduction of output signal range with the adjustment of the pull-down resistor, but the increase in valve size led to notably slower switching times than in their microfluidic counterparts. Subsequent designs incorporating materials like buckling sheets (45) or drinking straws (46) simplified manufacturing of the valves while bringing oscillator frequencies up to 15 Hz, albeit again at the expense of using pull-down resistors that reduce the output signal range. A more recent fluidic valve design enabled not only nearly full output signal range in an oscillator reaching 3.6 Hz but also analog pressure control in response to an analog input force (47). A valve design with a larger gate area than channel area allows amplification of pressure from the gate input to the actuator output for high fluidic power loads like soft actuators (48).

Despite these advancements, control in existing soft robots, particularly those that rely on fluidic logic circuits, is typically simple and slow when compared to commercial robotic systems, often driven by open-loop oscillating or undulating motions with timescales on the order of seconds or longer. Reducing the internal volume of soft actuators can bring their actuation time down into the range of 50 ms (49), but this work still requires an external electro-mechanical control valve with a sufficiently high flow rate and fast response time to drive the actuator. While the simplicity of control is often lauded as a selling point for using soft robots, for example, allowing grippers to use their inherent compliance to grasp objects with complex geometries without performing any computation, it limits the potential for applications that require more complex tasks and the ability to react to external events. Even fluidically controlled robots that have demonstrated tasks as complex as walking along a target path (50), bending to a desired curvature (51), or beating a video game (52) typically rely on simple open-loop underlying control schemes, such as a series of tuned oscillators, built-in hysteresis (53), or a preprogrammed external pressure regulation sequence, where the task has been carefully designed to fit to the robot's unique capabilities. While closing the control loop complicates system design, the addition of a feedback controller can enable disturbance rejection, reduce error from model uncertainties, reduce sensitivity to parameter variations, and improve stability and tracking performance. Recent work demonstrated closed-loop control of a soft actuator using a stretchable an electrical resistive strain sensor (54), but this setup relied on an array of external electronics and hardware infrastructure including an off-board pressure sensor, pressure regulator, and Arduino microcontroller.

As soft robotics research enters its second decade of prominence in academia, there is a desire to move beyond demonstrations that are “soft for soft's sake” and toward work that advances the state of the art both within and beyond the field, using quantitative benchmarks with contributions to robotics and engineering more broadly (55). To truly venture beyond the applications of soft grippers and oscillatory motions while operating within the fluidic domain, the field will likely require vastly more complex fluidic circuitry. This will demand several improvements to fluidic logic gates, including fast switching speeds, high reliability to enable larger circuits, and high output flow rates in dense packages. Toward this end, we present a versatile four-terminal fluidic transistor optimized for switching speed that consists of an injection-molded silicone rubber piston inserted into a three-dimensionally (3D) printed casing that

contains the four fluidic ports. We quantify its performance across a variety of metrics and compare it to the state-of-the-art where existing literature permits. The versatility of the four-terminal device to act as a two-way or three-way switch enables us to reduce the component count of existing digital logic circuits while adding capabilities like analog-to-digital conversion and analog flow amplification. Last, we demonstrate the value across domains by combining integrated subcircuits into a row-column decode circuit to drive a soft haptic bubble display for virtual reality, actuated with refresh rates faster than the human threshold for haptic latency perception (56), and a closed-loop controller that can drive a soft actuator to hold a target position in the presence of external environmental disturbances without using any electronics. In performing closed-loop position control with disturbance rejection, these circuits measure a dynamic input from the analog world, convert it to the digital fluidic domain to perform computation in a feedback controller, and use the result to drive an analog response at the actuator. The circuits accomplish this all in a layout that is scalable to higher resolutions and faster switching speeds, opening the door to advanced fluidic control of soft robots.

RESULTS

Digital logic with a four-terminal fluidic transistor

The fluidic transistor that serves as the fundamental building block in this work consists of an injection-molded silicone rubber piston resembling a poppet valve that moves within the stationary 3D printed casing with embedded fluidic routing that connects to each piston in four locations (Fig. 1A). The two layers of the casing hold the thin outer webbing of the upper disk of the piston to create a complete seal between the gate region and the flow regions. A taper on the tip of the piston's lower disk enables insertion through the column of the lower casing layer during assembly. The piston has two sealing surfaces within the flow channels: the upper sealing surface on the underside of the upper disk and the lower sealing surface on the upper side of the lower disk. The gate port connects directly to the gate region, the upper port connects to the underside of the upper disk, the output port connects to the cavity that surrounds the central column of the piston, and the base port connects to the underside of the lower disk of the piston (Fig. 1B). For digital logic, the device functions as a three-way, fluid-piloted switch such that, when the gate pressure is low, the flow channel between the base and output ports is open and the upper port is blocked by the upper sealing surface, and, when the gate pressure is high, the flow channel between the upper and output ports is open and the base port is blocked by the lower sealing surface. For the experiments in this work, we set the low pressure to be atmospheric pressure and refer any higher pressure by its gauge pressure. To test individual devices, we glue hypodermic tubing directly into the casing ports. To build integrated circuits, the casing layers include repeated patterns of empty spaces into which multiple copies of the injection molded pistons fit, with additional channels connecting the ports of the fluidic transistors to each other or to inputs and outputs of the circuit (Fig. 1C).

In the buffer configuration (Fig. 1D), the base port is held at a high pressure and the upper port is held at a low pressure. When the gate pressure is low, high pressure at the base is blocked from the output, which drains out of the upper port. When the gate pressure is high, the piston moves down to block the upper port and the base port pressurizes the output. Thus, in the terms of digital logic, the

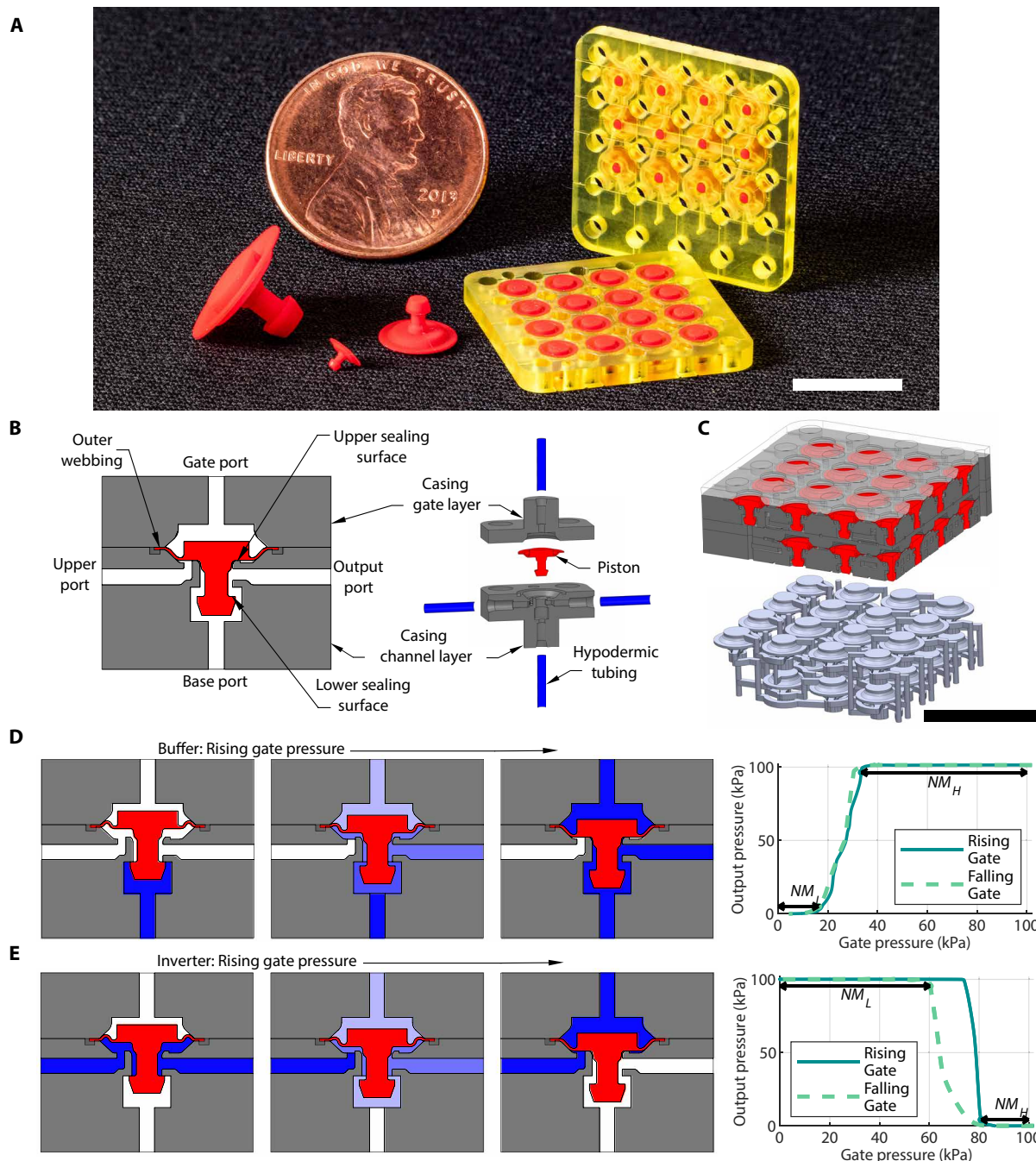


Fig. 1. The four-terminal fluidic transistor. (A) The three sizes of silicone pistons tested in this work with a penny coin (US\$0.01) for scale, alongside 28 of the 3.25-mm pistons installed into the two layers of casing and routing of a 4-bit full adder circuit. Scale bar, 1 cm. (B) A diagram indicates the port names, and an exploded view shows how an individual device is assembled for benchtop testing. (C) A cutaway view of the integrated 4-bit full adder and an inverted visualization of the negative space inside the casing show the internal routing of the assembled circuit. Scale bar, 1 cm. (D) In the buffer configuration of the device, the base port connects to a high pressure (blue) source while the upper port vents to the atmosphere (white) so that the output port pressure follows the gate port input pressure. (E) In the inverter configuration, the upper port connects to a high pressure source while the base port vents to the atmosphere so that the output port pressure inverts the gate port input pressure.

output pressure matches the gate pressure, while the piston acts to buffer the gate signal source from any loads on the output. Conversely, in the inverter configuration (Fig. 1E), the upper port is held at a high pressure while the base port is held at a low pressure, resulting in an output pressure that is high when the gate is low, and

low when the gate is high. Slow sweeps of the gate pressure while monitoring the output pressure provide insight into the device transients for both the inverter and buffer configurations in what we call the pressure transfer curve (PTC), analogous to a voltage transfer curve in electronics.

For cascable logic, the devices must switch at a gate pressure in between the high- or low-pressure rails. We calculate the high noise margin, NM_H , as the difference between the high source pressure and the gate pressure at which the output signal passes the 95% threshold of its range the gate pressure is rising, and the low noise margin, NM_L , as the gate pressure at which the output signal passes the 95% threshold of its signal range when the gate pressure is falling. We developed an analytical model (fig. S1) of the piston to better understand and optimize the device performance (derived in the Supplementary Materials). The device behavior during the transition between the pressure rails is nonlinear and hysteretic due to the changing internal flow resistances near the two sealing surfaces and the changing piston areas exposed to each of the pressures at the control ports when seals are made or broken. The inverter configuration exhibits greater hysteresis than the buffer configuration due to the large area on the underside of the piston's upper disk exposed directly to the source pressure when the piston is up. As a result, the inverter configuration requires a greater gate pressure than the buffer configuration for the piston to start moving down, and, once the seal at the lower disk is broken, the internal pressure drops and the high gate pressure quickly snaps the piston into its down position. From this position with the piston down and reduced pressure against the underside of the upper disk, the gate pressure must fall substantially before the piston will rise back up, creating hysteresis. Conversely, in the buffer configuration, once the gate pressure rises enough to break the seal at the lower disk, the pressure at the underside of the upper disk starts to rise and counteract the force exerted by the gate pressure, leading to a more gradual motion downward through its cycle that behaves more symmetrically when the gate pressure rises and falls than the snap-through behavior exhibited by the inverter.

On the basis of behavior exhibited in the PTCs and virtual design iterations with the analytical model, three key design features helped maximize the noise margin in these devices. First, the upper disk of the piston is larger than the lower disk of the piston, ensuring that there is always a larger area exposed to the gate pressure than there is exposed to the pressure in the flow channel regions. Second, the column of the piston has a slight knee outward at the top as it approaches the upper disk of the piston. We designed this knee in such a way to avoid constricting the flow path from the output to the upper port when the piston is all the way up, but to start constricting the upper flow path earlier in the piston's downward stroke than it would be if the knee did not exist. This leads to a pressure drop increasing earlier in the transition, allowing the downward force from the gate pressure to overcome the counteracting upward forces more easily, particularly in the buffer configuration. Third, the webbing of the piston is molded with the piston in its extended state so that the material buckles rather than stretches through most of its range of motion, reducing the force needed from the gate pressure needed to move it.

We tested three sizes of pistons that were scaled equally across all dimensions with two webbing preload directions (fig. S2) and six durometers of silicone to validate the scaling relationships of these devices. We denote the piston sizes of 3.25, 6.5, and 11 mm by their largest dimension, which is the outer diameter of the thin webbing that is sandwiched between the casing layers to isolate the gate region. The flow channel diameters for these three sizes are 425, 850, and 1450 μm , respectively. The silicone durometers range from 30 to 80 Shore A hardness in even increments. The resulting PTCs for source pressures of 100 and 200 kPa show how noise margins shift with piston durometer and size (fig. S3). We use the 3.25-mm, 60A

durometer, downward-preload piston for its optimal noise margin at 100-kPa source pressure for the remainder of this work, unless denoted otherwise. We cycle tested these pistons up to 5.5 billion cycles without failure (fig. S4), as explained in more detail in the Supplementary Materials.

In addition to functioning as buffers and inverters in a three-way valve configuration, these devices can act as two-way normally open or normally closed valves by blocking either the base port or the upper port, respectively. From a circuitry perspective, the four-terminal fluidic transistor acts as a single-device multiplexer, where the select input at the gate multiplexes the output between the base and upper port inputs. Following this logic, we can build two of the remaining basic logic gates (AND and OR) with just a single fluidic transistor and the remaining four gates (NAND, NOR, XOR, and XNOR) with just a pair of fluidic transistors (Fig. 2A), which we validate by building a full adder circuit (Fig. 2B) that functions with only seven fluidic transistors (Fig. 2C).

A kilohertz fluidic ring oscillator

In addition to optimizing the fluidic transistor design for noise margin in digital circuits, we have also optimized the design to reduce switching time. Prior work on fluidic ring oscillators relied extensively on long tubes in their designs, either as pull-down resistors or as part of the fluidic valves themselves (42, 44–46). As a result, the viscous losses in these flow channels dominate the inertial losses, resulting in laminar (low Reynolds number) flow. By minimizing flow channel lengths in our design, we not only reduce viscous losses but also decrease device size and internal capacitance, resulting in faster oscillators. However, as viscous losses decrease, flow transitions into turbulent (high Reynolds number) regimes and the effects of compressibility and choking become important to consider as Mach numbers increase. For laminar, incompressible flow, the Darcy-Weisbach equation (57) leads to a linear relation between pressure drop, ΔP , and mass flow rate, $\dot{m}$

$$R = \frac{\Delta P}{\dot{m}} = \frac{128\mu L}{\pi \rho D^4} \quad (1)$$

where μ is the dynamic viscosity, L is the pipe length, ρ is the density, and D is the pipe diameter. By comparison, the lumped parameter RC models for turbulent, compressible flow and deformable capacitors are nonlinear and derived in prior work (58). The resulting equation for turbulent, compressible flow resistance, $R^\dagger$, in an orifice, defined as a flow channel with a length less than approximately 40 times its diameter, is given as

$$R^\dagger = \frac{\sqrt{P_1(P_h - P_1)}}{\dot{m}} \text{ for } \frac{1}{2}P_h \leq P_1 \leq P_h \text{ (not choked) and} \quad (2)$$

$$R^\dagger = \frac{P_h}{2\dot{m}} \text{ for } 0 \leq P_1 \leq \frac{1}{2}P_h \text{ (choked)} \quad (3)$$

where the high and low pressures, P_h and P_l , respectively, are absolute pressures, not gauge pressures. The corresponding equation for compressible, adiabatic capacitance is given as

$$C = \frac{\rho_0}{\gamma P_0} \left(V_0 + V_m + \gamma P_c \frac{dV_m}{dP_c} \right) \quad (4)$$



inversely with the square of the flow channel diameter rather than the fourth power shown in the laminar flow models. We measure $R^\dagger$ values for our 3.25-, 6.5-, and 11-mm pistons' flow channels of 2.14×10^9 Pa-s/kg, 7.20×10^8 Pa-s/kg, and 2.30×10^8 Pa-s/kg, respectively, using the methods outlined in our previous work (55). These correspond to mass flow rates of 3.77×10^{-5} kg/s [1.75 Standard Liters Per Minute (SLPM)], 1.29×10^{-4} kg/s (6.01 SLPM), and 4.00×10^{-4} kg/s (18.5 SLPM), respectively, at 100 kPa. Given that the volumetric flow rate, Q , is

$$Q = \frac{\dot{m}}{\rho} = A\bar{v} \quad (7)$$

where A is the cross-sectional area of the flow channel and $\bar{v}$ is the average velocity in the channel, we calculate the Reynolds number, Re , as

$$Re = \frac{\rho\bar{v}D}{\mu} = \frac{4\dot{m}}{\pi D\mu} \quad (8)$$

This yields Reynolds numbers of 6.24×10^3 , 1.07×10^4 , and 1.94×10^4 , respectively, for the increasing piston sizes, all in the turbulent regime. Gate volume essentially dictates the capacitance, which thus scales with the cube of the length dimension. As a result, the switching time for a fluidic transistor to drive a single fan-out in a ring oscillator scales linearly with the length dimension of the overall design.

We chain three of our 3.25-mm, 60A fluidic inverters into a ring oscillator (Fig. 3A) with a pressure sensor attached to one output to demonstrate rail-to-rail signal propagation with a cycle period of 1.6 ms (Fig. 3B) and a switching frequency of 629 Hz for each inverter. Performing a fast Fourier transform (FFT) on both the output pressure data and the audio data from the recording of the test shows a clear harmonic with a fundamental frequency at the oscillator frequency of 629 Hz and a peak at the third harmonic of 1887 Hz (Fig. 3C). Each of the three fluidic transistors switches at the fundamental frequency, and, with three transistors in each oscillator, the FFT power spectrum shows a clear peak at the third harmonic. To validate our scaling laws, we need to remove the pressure sensor because the load of its fixed dead volume capacitance does not scale with the fluidic transistor volume and thus slows the smaller piston oscillators more than it slows the large piston oscillators. After removing the capacitance of the pressure sensor, the audio data show that the fundamental frequency of the 3.25-mm, 60A oscillator at 100-kPa source pressure rises to 701 Hz. We tested source pressure ranging from 25 to 200 kPa in 25-kPa increments for all our piston sizes and durometers, resulting in fundamental frequencies in the audio data ranging from 140 to 1017 Hz (Fig. 3D). We include a video of the oscillator test for the 3.25-mm 60A durometer piston that is used for the digital logic in this work (movie S1) where the oscillation frequencies are clearly audible, and we also include the audio recordings of all oscillator tests in the Supplementary Materials. The data for the 60A durometer are isolated to show the linear scaling relationship with piston length scale (Fig. 3E), as predicted by our response time models.

Our scaling analysis and experimental results suggest that increasing the size of a fluidic valve design to achieve higher output flow rates should result in slower oscillations, resulting in a trade-off between the two desired metrics. To place our work more clearly in the context of the existing literature, we created an Ashby chart for fluidic ring oscillators, plotting oscillator frequency as a function of output flow channel dimension (Fig. 4). We use output flow channel

dimension rather than output flow rate on the x axis to eliminate the dependence of the flow rate on the source pressure and to account for the fact that many existing papers do not list output flow rates or clear flow resistance metrics.

A low-latency soft haptic display for virtual reality

The fast response time and relatively high output flow rates of the fluidic transistors developed in this work enable applications of fluidic multiplexing for which microfluidic valves are too small and existing macroscale valves are too slow. For example, a soft haptic bubble display could be useful for rendering the surface textures or shapes of objects in a virtual environment as the user interacts with the virtual objects dynamically. A simple way to control such a demo is to connect one electromechanical valve to each haptic bubble display to drive each pneumatic channel directly, but, for high-resolution displays, this results in many bulky electromechanical valves that are unlikely to be mobile or wearable. Various schemes for fluidic multiplexing or shift registers can reduce the number of electromechanical valves by converting serial signals from the electromechanical valves into parallel outputs but at the expense of increasing the number of switching steps required. This creates a trade-off between the number of electromechanical valves required and the frequency at which the entire system must be able to actuate. By reducing the response time of the fluidic valves, we can effectively increase the number of switching steps that can fit into the 50-ms window of latency that is below the threshold for human haptic perception (53) such that pneumatic bubbles in an array could be used in a haptic display for virtual reality. Such an array worn in a glove could simulate making and breaking contact with a virtual object or dragging a finger across an uneven surface like a virtual keyboard by selectively pressurizing against the hand in the locations where the virtual object touches the hand.

The fluidic circuit that we use to drive the soft haptic display consists of a pattern of three fluidic transistors (Fig. 5A), one as an AND gate and two as a buffered latch to hold the state, repeated for each of the $m \times n$ outputs in a square array (Fig. 5B). When a given output's row and column inputs are both high, the latch releases to connect the drive input to its bubble display output. If either of its row or column inputs is low, then the buffered latch holds its state. Thus, we can control each of the outputs independently in a maximum of m or n switching cycles. The buffered latch ensures that the system is tolerant to some leakage in the interfaces between the circuit, the tubing, and the bubbles, provided that the leakage rate does not eclipse the valve flow rate. To find the minimum switching time for a decode circuit element, we reduce the time step between commands on the electromechanical valve inputs until the output signal is compromised. A 50-ms switching time produces a clean digital output signal, while a 10-ms switching time remains functional but shows the RC dynamics of the circuit on the output signal. At a 5-ms switching time, both the RC dynamics and the latency from the electromechanical valve are clear in the output data, and, at 2 ms, the electromechanical valve latency dominates the switching speed, and the RC dynamics of the fluidic transistors prevent the circuit from fully functioning (Fig. 5C). We use 13 total electromechanical inputs from a piezo valve array to drive each of the six row and six column inputs into the circuit as well as a single drive input that connects to all the buffered latches. We show a range of output sequences (Fig. 6 and movie S2) demonstrating independent control of each output in real time.

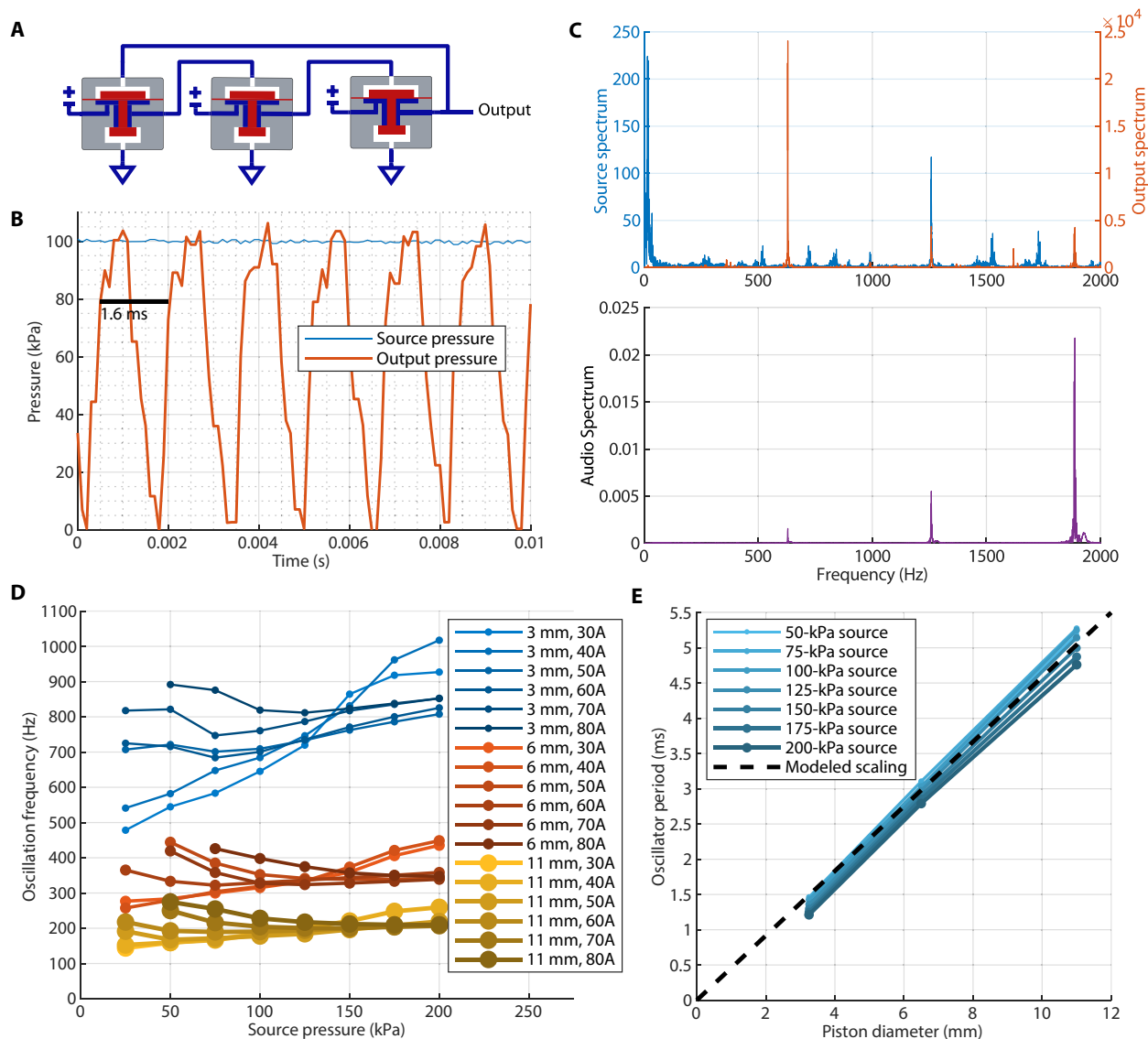


Fig. 3. Ring oscillators using three inverters. (A) A circuit diagram with four-terminal fluidic transistors. (B) With a pressure sensor on one output, a ring oscillator with 3.25-mm 60A pistons oscillates with full signal propagation in a period of 1.6 ms. (C) Audio analysis of the oscillators shows a clear harmonic at the oscillation frequency, allowing removal of the pressure sensor to test scaling across device size without a load. (D) Frequency data from three piston sizes and six durometers across a range of source pressures. (E) The oscillator period scales linearly with device size as predicted by analytical RC models.

A fluidic analog-to-digital converter and push-pull amplifier

While the results presented above focus on digital fluidic logic, robots are ultimately operating in an analog world and need a way to transition between the two domains to use digital logic for computation and robust control. Several relatively straightforward passive circuits allow the conversion of a set of digital outputs to an analog output, such as a resistor string digital-to-analog converter (DAC) or a resistor ladder DAC. Pulse-width modulation (PWM) requires active components to drive an analog signal with digital circuits but can allow greater efficiency and precision than passive resistor DACs. We demonstrate a 3-bit PWM circuit using 22 of our four-terminal fluidic transistors (fig. S5) in the Supplementary Materials. An analog-to-digital converter (ADC), on the other hand, typically relies on comparators, requiring a sharp and consistent transition to

ensure a reliable digital output. Although the PTC characteristics of our four-terminal devices lack this sharp transition on the output for a sweep of the gate input, by venting the upper port and output port to ambient pressure and adding a pull-up resistor to the base port (Fig. 7A), we can generate a very sharp transition using the pistons as three-terminal, normally closed devices (Fig. 7B). For the 3.25-mm 60A piston with a 150- μ m orifice acting as the pull-up resistor, this transition occurs at a gate pressure of 20 kPa for a source pressure of 100 kPa with about 4 kPa of signal lost on the low end from the pull-up resistor. A larger pull-up resistor, from a smaller orifice diameter, lowers the output pressure closer to the low-pressure rail when the valve is open and consumes lower steady-state flow but at the expense of a slower response time for the output pressure to rise when the valve closes. To raise the switching

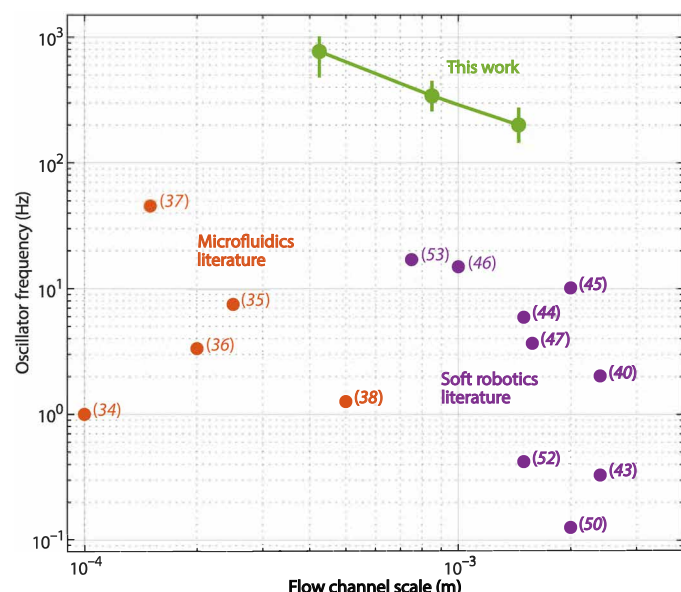


Fig. 4. Device switching speed in the context of previous research across domains. Scaling analysis suggests that fluidic oscillators become slower as overall device size increases. The four-terminal device in this work results in orders of magnitude faster oscillations than previous research while maintaining flow channels large enough to drive soft robotic actuators.

threshold to a higher gate pressure, we can add a pair of resistors to create a pressure divider between the analog signal and each gate, and to lower the switching threshold we use a pressure divider to reduce the pressure at the base port of the transistor. For a 3-bit ADC circuit (Fig. 7C), we use one piston with a reduced switching threshold and six pistons with an increased switching threshold such that the seven transitions happen at roughly equally spaced intervals along the analog input sweep. We add a four-terminal inverter to each output to reverse the inversion of the signal created by the three-terminal device with the pull-up resistor (Fig. 7D). The manufacturing tolerance of our 3D printed orifice resistors limits the precision of the ADC switching intervals. For applications in which the sensor output does not span the full signal range, the ADC input resistances can be adjusted accordingly to maintain full usage of the resolution and remove the need for any pistons with reduced switching thresholds.

This ADC topology presents two major challenges. First, there is a large amount of hysteresis in its performance as each of the switching thresholds are very close to zero when the gate pressure is falling. This means that, while we can reliably convert the analog signal to the digital domain while the signal is rising in pressure, the outputs will be stuck high until the signal falls all the way back to nearly atmospheric pressure. To resolve this, we address the problem by clocking the ADC with a ring oscillator and flip-flops in the digital domain as described in the following section. The second problem, which requires a unique fluidic circuit to address, is that the pressure dividers used to adjust the switching thresholds of the ADC draw a constant flow from analog signal source, which can affect the pressure level of the signal itself. In electronics, an operational amplifier can buffer the analog signal without drawing any current at its inputs, but our single-piston devices were originally designed to buffer only digital signals.

To address this second challenge, we combine a pair of pistons configured as three-terminal, normally closed two-way valves in series to act as a push-pull amplifier by adjusting the casing dimensions for the second device so that the underside of the lower disk of its piston acts as the sealing surface rather than the top side of the lower disk (Fig. 8A). An increase in the gate pressure relative to the output creates a force imbalance across the top disks of the pistons, pushing them downward and opening a flow path to the source pressure on the first piston while the second piston remains closed. A decrease in the gate pressure relative to the output creates a force imbalance across the top disks of the pistons that pushes them upward, holding a seal on the first piston while opening a flow path to the atmosphere on the second piston. The two pistons work in tandem to reduce any discrepancy between the target input pressure at their gates and the output pressure (Fig. 8B) while doing so in such a manner that the two flow paths are never open simultaneously, preventing steady-state flow through the amplifier or from the target input into its gates. The geometry of the piston does create some distortion on the output pressure signal relative to the input pressure.

Electronics-free closed-loop position control of soft actuators

At an abstract level, four fundamental components comprise a closed-loop controller: a plant (dynamic system under control), a sensor to measure the output of the plant, a feedback subtractor to calculate the error between a reference target input and the measured output, and a controller. With our four-terminal fluidic transistor, we now have the subcircuits needed to build all these system components without electronics (Fig. 9A). For this demonstration of closed-loop position control, the plant consists of an inverse pneumatic artificial muscle (59) that functions similarly to an expandable garden hose; a strain-resistant fabric or string surrounds a soft latex or silicone tubing, constraining its expansion in the radial direction such that it expands along its length when pressurized and contracts when depressurized. We use a smaller silicone tube constrained to stretch in parallel with the actuator to act as a pneumatic sensor. As the actuator and sensor tube stretch lengthwise, the sensor tube flow resistance increases both due to its length increasing and its diameter decreasing. We add a fixed resistance orifice connecting the source pressure to the sensor tube, creating a pressure divider such that the output pressure between the two flow resistances increases with the strain of the sensor tube, analogous to the functionality of a voltage divider circuit in electronics.

A push-pull amplifier buffers the analog sensor output before passing it to the input of the ADC so that the varying flow into the ADC input does not distort the accuracy of the sensor output. A fluidic priority encoder converts the eight levels of digital outputs from the ADC into a 3-bit binary output. To address the hysteresis of the ADC, we add a ring oscillator to clock the circuit. The clock signal simultaneously multiplexes the input of the ADC between the sensor output and atmospheric pressure while synchronously driving a set of three flip-flops in parallel on the priority encoder outputs. With each clock cycle, the ADC input drops to ambient pressure before rising to the current sensor pressure while the previous cycle's sensor reading is passed through the flip-flops. Thus, the ADC always reads the sensor output as a rising pressure, eliminating the hysteresis, while the flip-flops prevent the low output of the ADC between each sensor reading from passing through to their

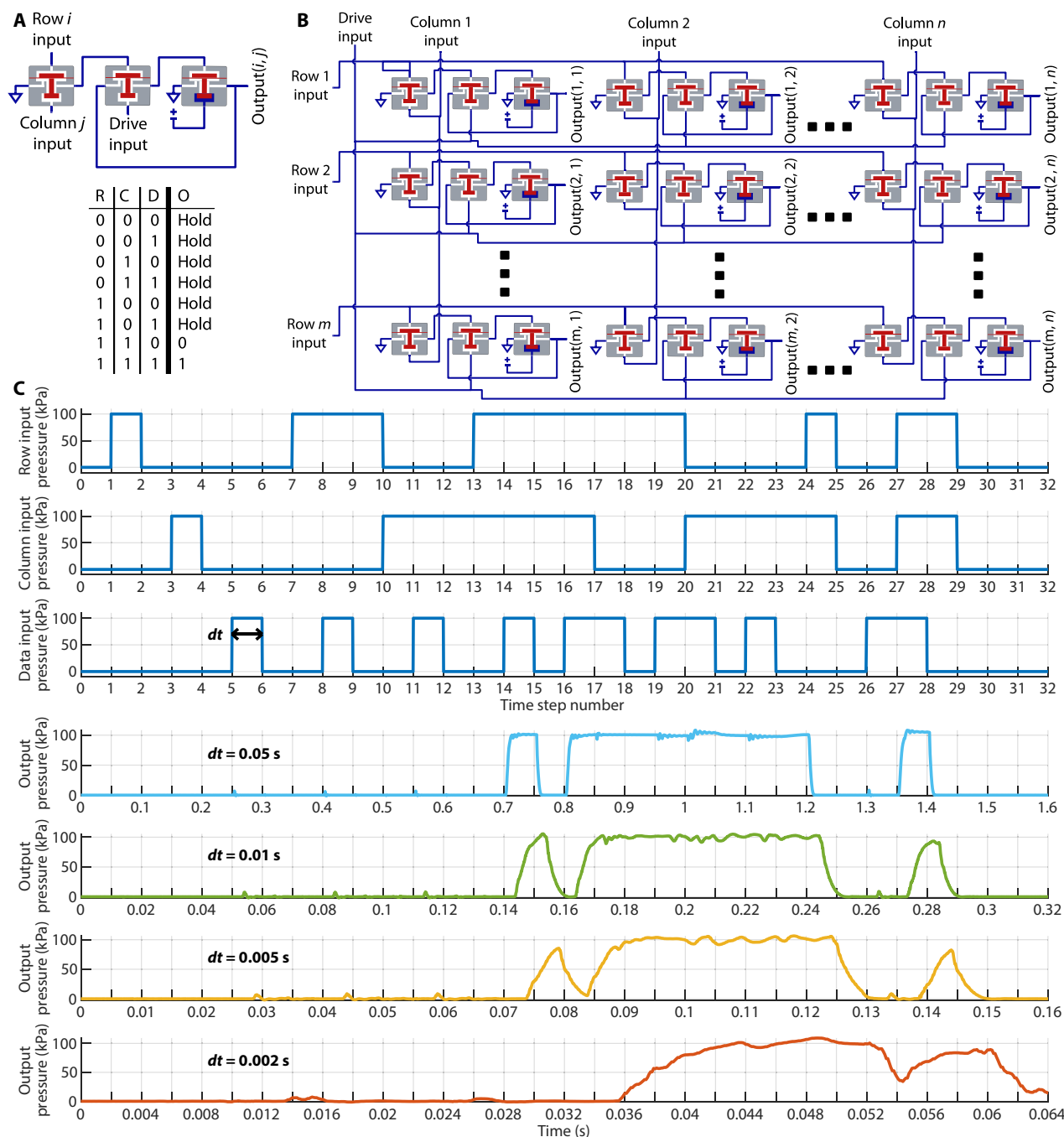


Fig. 5. A row-column decode circuit. (A) A circuit diagram of the row-column buffered latch. The data input connects to the output if both the row and column inputs are high; otherwise, the output is held and buffered. (B) The buffered latches are combined in m rows and n columns to control $m \times n$ outputs with only $m + n + 1$ inputs. (C) Data showing the performance of the latch at a range of switching speeds.

outputs. We prototype the full circuit on a large “fluidic breadboard” machined from aluminum that acts as a manifold routing the single source pressure line to each of the subcircuits. The parasitic capacitance from the tubing between subcircuits substantially slows the propagation of digital logic around the control loop, necessitating the addition of a 5-ml syringe to the oscillator to slow down the clock. Silicone bubbles display intermediate pressure levels, akin to LEDs on an electrical circuit board.

To calculate the error between 3-bit reference target input and 3-bit measured position, we use a 4-bit full adder circuit as a subtractor with ones’ complement notation. We invert the 3 bits of the measured position input into the adder and set its fourth, most significant, bit permanently high as a negative sign. Ones’ complement subtraction creates an off-by-one error when the result is positive, but, conveniently, the carryout value of the highest bit will always be high when the result is positive, so we can feed this carryout output

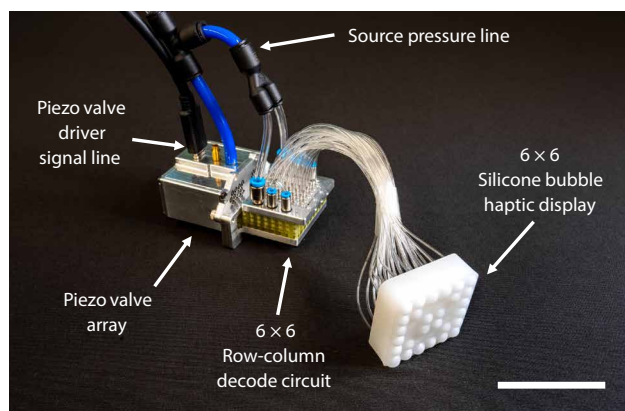


Fig. 6. A low-latency soft haptic display for virtual reality. A sample output of a six-by-six decode circuit controlled by 13 piezo valves on a soft haptic bubble display. Scale bar, 5 cm.

back into the carry-in input on the lowest bit to fix this off-by-one error. The fourth, most significant, bit of the output, which represents a negative calculated error, multiplexes between inverting and buffering the first 3 bits to convert them into a positive binary value for the control magnitude. The fourth bit also multiplexes the controller input between the high source pressure and the low

atmospheric pressure so that a negative calculated error, or under-actuation, connects the actuator to high pressure to further extend the actuator, while a positive calculated error, or over-actuation, connects the actuator to atmospheric pressure to contract it. The three bits that denote control magnitude connect, in order of increasing bit significance, to one, two, and four fluidic transistors, respectively, each routed in parallel as two-way normally closed valves between the control multiplexer (the direction of which was set by the fourth bit) and the actuator. In total, 90 four-terminal devices constitute the full fluidic feedback circuit from sensor to actuator (Fig. 9B). We include a video walking through the assembly of each of the subcircuits into the control loop, using bubble displays between each major component to demonstrate functionality (movie S3) along with the full circuit diagram (fig. S6) in the Supplementary Materials.

Because the mass of air inside the actuator determines its pressure and, therefore, its position, and the controller sets the mass flow rate into or out of the actuator as a linear multiple of the calculated error, the overall scheme functions as an integral controller. To demonstrate the performance of the controller and capabilities for disturbance rejection, we include a video of the actuator reacting to dynamic events like step-change increases or decreases to the weight on its end effector (movie S4). We track the position of the end effector in each frame and compare it to the response of the same actuator with an open-loop controller consisting of a regulated analog pressure input (Fig. 10A) with a few select frames and

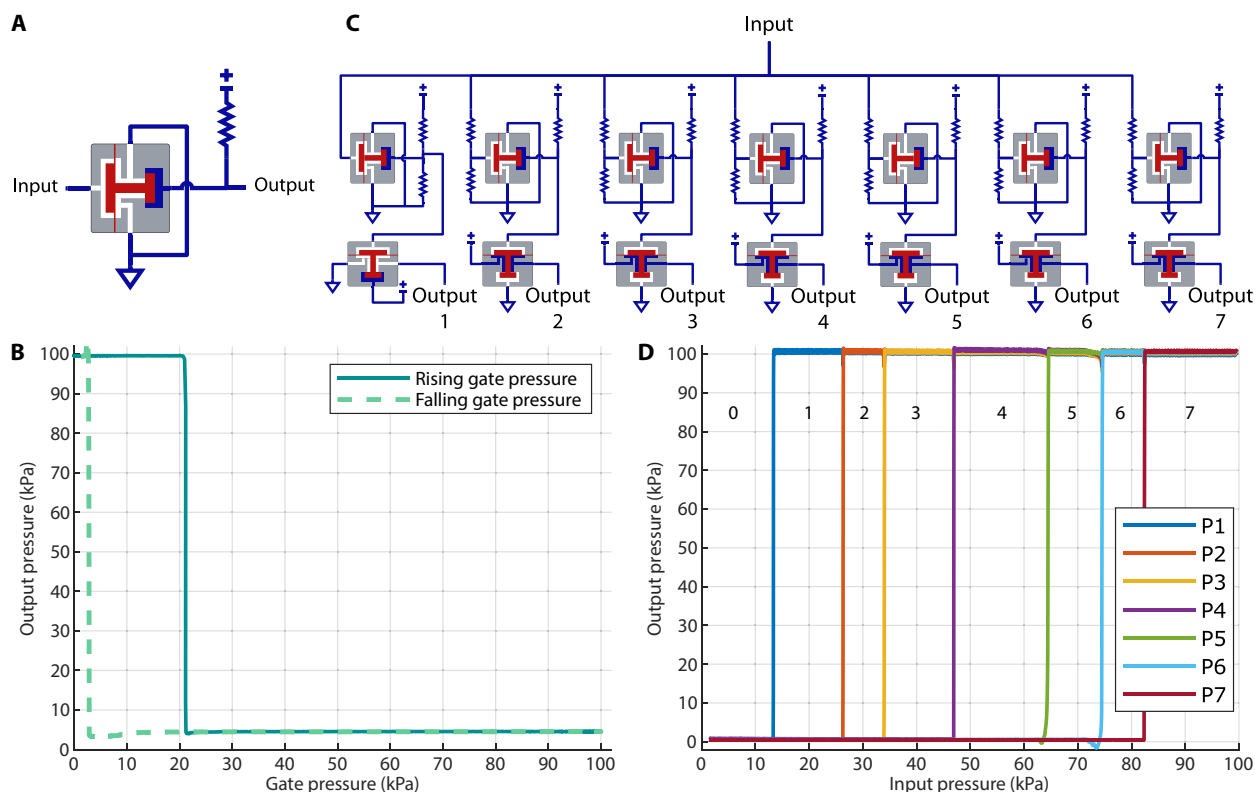


Fig. 7. Analog-to-digital conversion and analog amplification with fluidic transistors. (A) Venting the upper port of the four-terminal transistor creates a normally closed two-way valve. (B) Adding a pull-up resistor on the output creates a sharp transition on the PTC using resistor-transistor logic. (C) Pressure dividers allow tuning of the switching thresholds of the two-way valves so that multiple valves create an ADC. An additional inverter on each output reverts the signal to rises to rise with rising input pressure. (D) Seven switching thresholds create eight levels of digital outputs for a 3-bit ADC with inverters on each output.

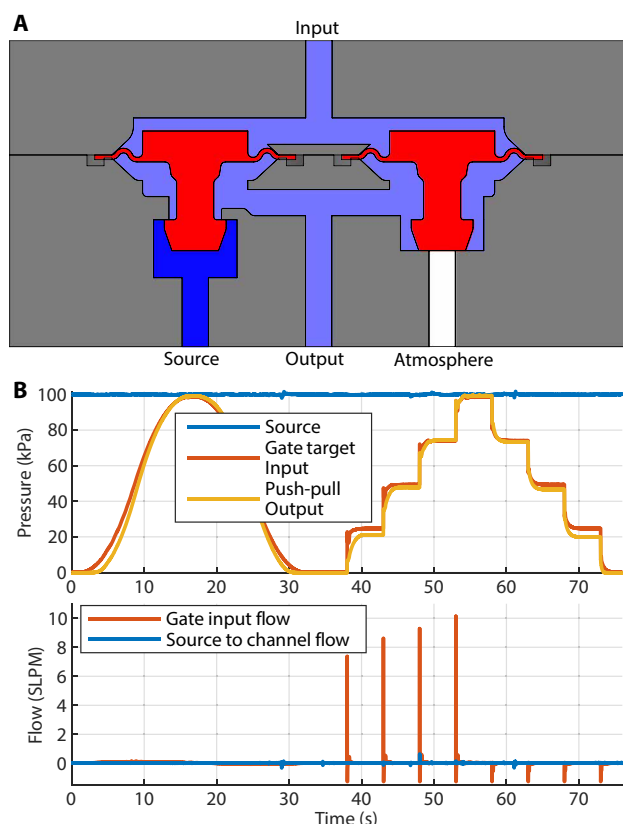


Fig. 8. Analog flow amplification and signal buffering with fluidic transistors. (A) A two-piston push-pull amplifier created by adjusting the casing dimensions around the second piston. (B) The push-pull amplifier passes the analog input signal to the output without drawing any steady-state flow.

corresponding end effector positions highlighted (Fig. 10B). While the position of the end effector with the open-loop controller oscillates until settling at a position several centimeters away from the initial target position, the closed-loop controller damps the oscillations and continuously adjust the pressure in the actuator until the resting position falls within the sensing resolution of the 3-bit ADC, resulting in only a few millimeters of change in position.

DISCUSSION

The four-terminal fluidic transistor presented in this work shows a high versatility for use within digital circuits, analog circuits, as well as circuits bridging the domains in both directions from analog sensors to digital computation and back to control of analog actuators. It achieves this with low cost, high reliability, and fast switching speed. In addition to demonstrating billions of life cycles while maintaining functionality, the four-terminal device reduces the number of fluidic transistors required to build basic digital logic gates and subcircuits, helping address the “tyranny of numbers” problem (30) with large, complex circuits. For example, the 3-bit closed-loop position control circuit presented in this work used 90 four-terminal devices with one moving component each, whereas, with traditional three-terminal devices, the same circuit would require at least 180 active devices with at least as many moving parts.

Using a hypothetical 95% yield, the circuit with four-terminal devices would function with more than 100 times higher probability than if it were designed with three-terminal devices of the same yield, although either case would likely require the addition of some fault tolerant logic or rigorous quality control.

The ring oscillator frequencies presented in this work represent a 20-fold increase when compared to the fastest oscillators found in the microfluidics literature, a 70-fold increase when compared to the fastest oscillators found in the robotics literature, and a nearly 300-fold increase over the fastest oscillators that use complementary logic to enable the full signal range on the output, all while achieving output flow rates on the order of liters per minute as needed to drive many soft robotic applications. Lower durometer pistons show a larger dependence in oscillator frequency on source pressure than higher durometer pistons, with the softest pistons at high source pressures showing the highest frequencies, although these source pressures are close to their limit of functionality. Some of the lowest durometer pistons failed to oscillate at the highest pressures tested, while some of the highest durometer pistons failed to oscillate at the lowest pressures tested, as we expected based on the noise margins measured in the quasistatic PTC test results. While we chose silicone durometers ranging from 30A to 80A to optimize performance at 100-kPa source pressure, the data suggest that, for softer pistons to function at higher source pressures of 200 kPa or greater, the center column of the piston would need to be designed with a larger diameter or, otherwise, reinforced to prevent it from stretching under pressure. Similarly, we would expect that an extra soft durometer silicone like Shore 10A or below Shore 00-50 might require some reinforcement of the center column to function at 100 kPa or greater but see no evidence that a softer piston would fail to function at lower source pressures even without design modifications. Given that the pistons tend to behave relatively stiffer as the design is scaled down, it is likely that an extra soft silicone would be preferable to further miniaturize the device any smaller than the 3.25-mm diameter tested in this work.

The results and analysis of this work show that linearly scaling all dimensions of the piston design down in size leads to a linear decrease in propagation delay of signals between transistors in logic circuits due to the quadratic increase in flow resistance and cubic decrease in gate and flow channel capacitance. Thus, smaller devices benefit fluidic computation circuits both for the increased number of components that can fit into a given package and for the increased clock speed at which such circuits can run. While the propagation delay between fluidic transistors does increase as the device design dimensions are scaled larger, these larger devices with their correspondingly larger output flow rates provide utility as power amplifiers. Even at these larger scales, the optimization of the fluidic transistor design for fast switching benefits the overall system when a power amplifier interfaces with smaller devices in a computation circuit. As with an electrical system that includes logic transistors within a microcontroller and power transistors to drive electromechanical loads, our closed-loop control demonstration uses the smaller 3.25-mm-diameter fluidic transistors to run the logic subcircuits and larger devices for flow amplifiers. Our 11-mm-diameter piston fluidic transistor has an output flow rate over 18 SLPM at 100 kPa to drive a soft actuator with high power while still oscillating at over 200 Hz with full signal propagation in a three-device ring oscillator, enabling it to respond in milliseconds to an input from another fluidic device. For any applications that require even larger

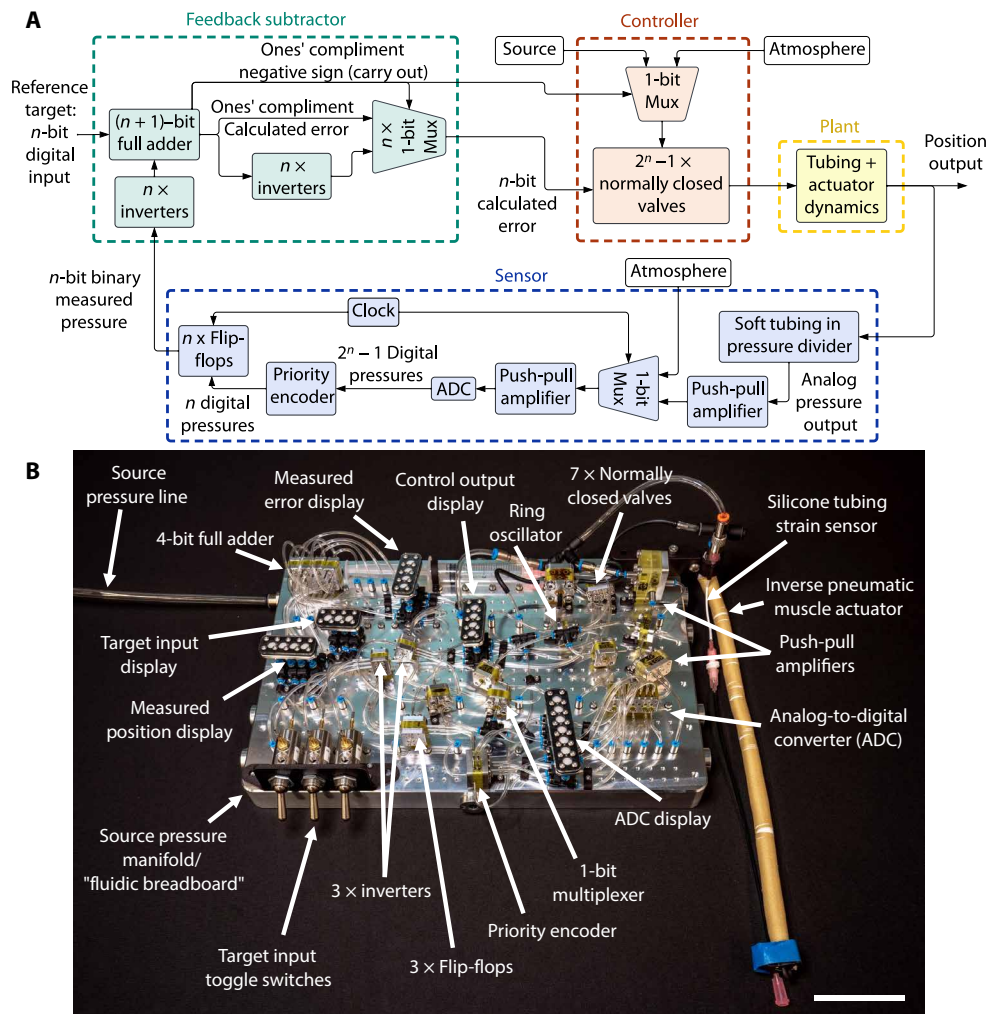


Fig. 9. Electronics-free closed-loop position control. (A) The four-terminal fluidic transistor enables all four key subcomponents of a feedback controller entirely within the fluidic domain. (B) A complete closed-loop fluidic position controller circuit assembled on a machined base that serves as the equivalent of a breadboard. Scale bar, 5 cm.

flow rates to drive actuators with more power, the fluidic transistor design could easily be scaled larger without any major changes to our manufacturing methods. In such a scenario, a chain of transistors with incremental size increases can reduce the latency compared to driving the large transistor gate directly with a small logic transistor, analogous to an “exponential horn” in electronics (60).

The volume of rigid components used in this work represents the biggest limitation of its potential application to soft robotics. Although we relied on 3D printing of the casing parts and rigid components like screws and needles to assemble and clamp the subcircuits demonstrated in this work for quick prototyping, future designs for manufacture at scale could use the same liquid silicone rubber (LSR) injection molding techniques that we used for the pistons to also manufacture the casing components, which could facilitate easier chemical or plasma bonding of silicone-silicone interfaces to improve the seals between circuit layers while enabling fully soft devices. In addition, building the closed-loop control circuit with discrete subcircuits and tubing on a fluidic breadboard, which added substantial parasitic capacitance between components,

severely limited the clock rate and potential control bandwidth of the circuit. Combining all the subcircuits into a single integrated printed circuit would markedly reduce the overall volume, and manufacturing improvements to reduce the size of individual fluidic transistors would further shrink the package volume while increasing computation speed. However, even with such advances we do not anticipate that a fluidic computation circuit would ever outperform a purely electrical control system for an application in which electronics could be easily integrated. Rather, the value of fluidic logic circuits remains in applications that benefit from remaining entirely within the fluidic domain or to remove electromechanical components from a system architecture.

A more thorough study and statistical analysis than the life cycle testing presented here would provide clearer insight into the viability of large-scale integrated circuits, but the early experiments in this work offer promising results for the durability of the devices. While the tooling for each piston design has a notable upfront cost at US\$3500, the unit price to injection mold each piston, including labor, machinery, and materials, is only US\$0.28, bringing the total

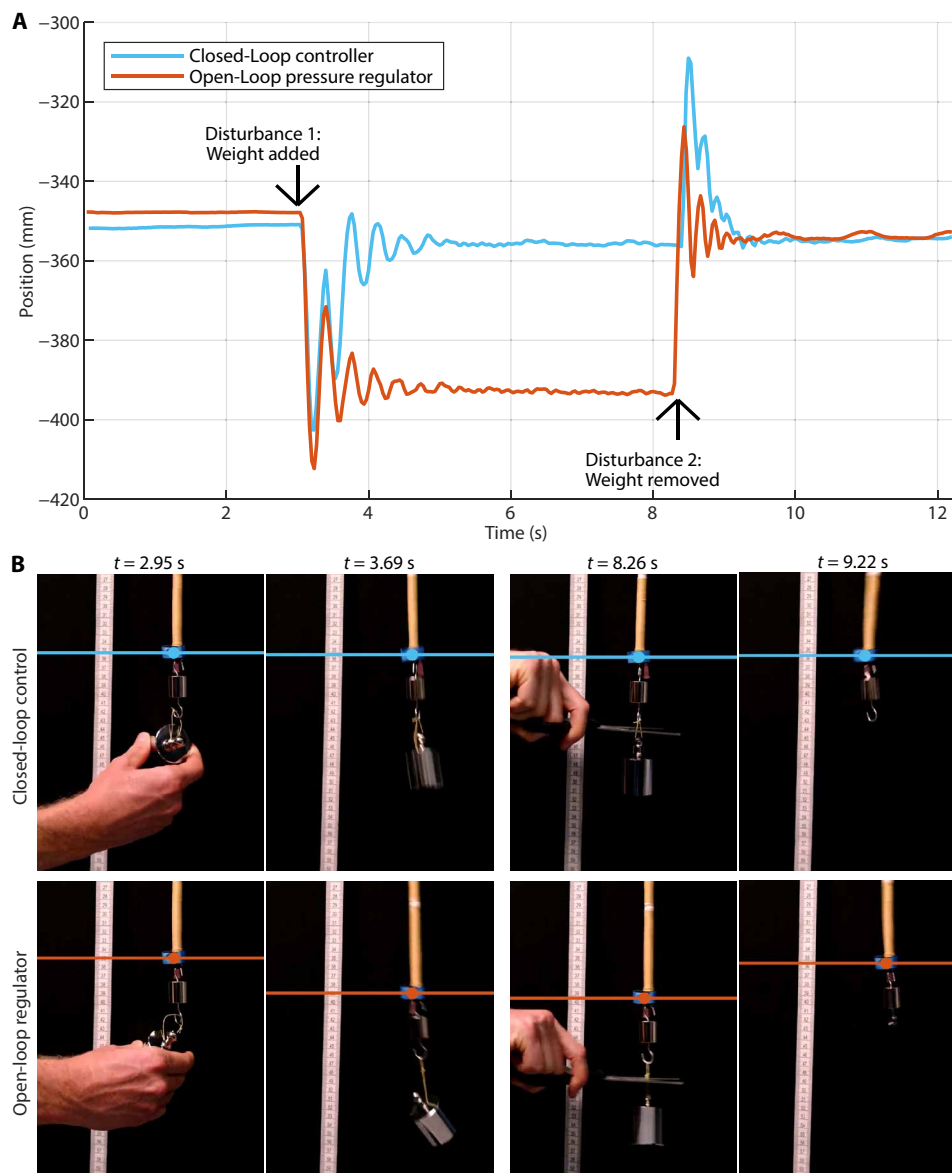


Fig. 10. Disturbance rejection in a fully fluidic position control circuit. (A) The feedback control circuit drives the end effector position back to the target within the resolution of the ADC while an open-loop regulated pressure input has no mechanism for disturbance rejection. (B) Frames captured from movie S4 that generated the data in (A).

cost at higher volumes (10^3 to 10^4 components) close to those touted in recent work (45, 46). This holds particularly true if one assigns a nonzero value to labor involved in assembling those more complex valves that have larger numbers of discrete components. Furthermore, the use of injection molding for the silicone pistons translates to mass manufacturing more readily than the soft lithographic techniques typically used in the microfluidics literature or bespoke processes typically used in the soft robotics literature. While lithographic techniques often require skilled use of a variety of specialized lab equipment and backend processes for surface modification, assembly, and bonding tailored to a specific design, a variety of commercial manufacturers offer LSR injection molding for both prototyping and mass manufacture, albeit at the expensive of larger minimum design feature dimensions. Although the devices here

approach the smaller end of the current limits for injection molding, the design could easily scale orders of magnitude larger using existing mass manufacturing techniques.

Several examples of previous work with fluidic multiplexing show a greater increase in the number of outputs per number of control inputs than our row-column decoder, but the purpose of this demonstration is to show the speed at which our fluidic transistors can switch in complex circuits as well as the utility of this improvement in switching speed. In the case of the soft haptic bubble array, a circuit built with slower switching speeds or lower output flow rates would not be capable of driving the display with the low latencies needed to stay below the threshold for human haptic perception. Similarly, while it is possible that a much simpler circuit could be constructed to perform fully fluidic position control of a

soft actuator like the one shown in this work, the aim of our demonstration is to show the feasibility of integrating all of these complex fluidic subcircuits to perform a dynamic control task representative of a vast number of robotic control tasks in the broader field, all within a circuit topology that can scale directly to greater bits of resolution as needed when crossing between digital and analog domains. Although our demonstration of closed-loop position control was limited to just a few loop cycles per second, we estimate that by combining the subcircuits into a layout more closely analogous to a printed circuit board than a breadboard, we could reduce the tube routing capacitance and latency by a factor of 20. Such a reduction in tube capacitance could bring the overall control loop frequency into the range of hundreds of cycles per second, approaching the cycle speeds used in many conventional robotics applications, still well within the frequency range the ring oscillators in this work demonstrated. By enabling electronics-free robots to not only sense and act in the pneumatic domain but also think, we hope to push soft robotic control beyond simple oscillations and binary state changes to a richer, dynamic interaction with the analog world.

MATERIALS AND METHODS

Objectives and design of the study

The objective of this study is to demonstrate a fluidic transistor design that can drive digital circuits and analog circuits and convert back and forth between digital and analog domains while functioning reliably at high frequencies with output flow rates high enough to dynamically control soft robotic applications. We use quasistatic analytical models to optimize transistor performance, build and test ring oscillators to quantify switching speeds, and run cycle tests to evaluate lifetime reliability. The soft haptic display for virtual reality and electronics-free position control of a soft robotic end effector with disturbance rejection provide practical examples for uses of the fluidic transistor.

Fabrication of samples

All the fluidic transistor pistons in this work were injection molded (Pioneer Material Precision Tech) using PMP-P2100-520 U 30A-80A LSR. All casing parts were 3D printed (Boston Micro Fabrication 10- μ m S240 printer) using BMF-TOUGH material. Prototype subcircuits were hand-assembled and clamped between aluminum machined layers with alignment features. To interface with individual devices, we use Loctite 4311 UV cure glue to attach thin wall stainless steel hypodermic tubing. Flexible thermoplastic polyurethane tubing (2-mm OD Festo 133038 PUN-H-2X0,4-NT or 4-mm OD Festo 197376 PUN-H-4X0,75-NT) easily pushes over these hypodermic tubes (18- or 14- gauge) to create a leak-free interface or into push-to-connect fittings (2-mm Festo 133026 QSM-M3-2-I or 4-mm Festo 133002 QSM-M3-4-I-R). To bond casing layers together after assembling all the pistons in a circuit, we laser cut Tesa 4983 double-sided acrylic adhesive to match the surface pattern on the casing to keep routing channels clear. However, such bonding prevents nondestructive disassembly of devices, so we used machine screws and a threaded aluminum cap to hold the layers together for most of the integrated circuits in this work. Computer-aided design (CAD) files for all circuit designs and individual pistons are included in the Supplementary Materials.

The piezo valve array to drive the decode circuit for the soft haptic bubble display consists of Johnson-Matthey piezo benders driven

by an Analog Devices LT3757 flyback converter with an FTDI FT4222HQ-D USB to I2C bridge to communicate with a PC. The CAD file for the soft haptic bubble array, included in the Supplementary Materials, consists of Spectroplast TrueSil A35 silicone 3D printed by Spectroplast AG on Silicone Additive Manufacturing machines. We press-fit 2-mm flexible tubing into the back of the haptic display that interfaces with hypodermic tubing on the outputs of the piezo valve array.

The fluidic breadboard consists of machined aluminum, and the Supplementary Materials includes the CAD and machining drawings. Threaded blanking plugs (Festo NPQU-BK-G14-P10) block any unused ports on the breadboard. Single-action air directional control valves (McMaster 8399 K13) allow manual changes to the pressure inputs. The soft actuator consists of a silicone tubing (McMaster 5054 K826) wrapped with a Kevlar thread (McMaster 8800 K44), and the strain sensor is a silicone tube (McMaster 5236 K501) with standard Luer lock fittings (McMaster 75165A676).

Testing methods

Our experimental setup used Honeywell ASDX Series pressure sensors to measure pressure at the inputs and outputs of circuits. Hoerbiger Pre-U2 pressure regulators precisely control pressures at the ports of the fluidic transistors in quasistatic PTC tests. A Honeywell Zephyr HAF series mass flow sensor measures mass flow rates through the devices in quasistatic tests. Enfield LS-V05 solenoid valves drive the inputs for circuit functionality tests. Data acquisition, regulator control, and valve control were synchronized through a National Instruments CompactRIO controller with a sample rate of 10 kHz for dynamic tests and 100 Hz for the quasistatic PTC tests and circuit functionality. Flow resistance tests and parameter fitting methods are explained in detail in our prior work (55).

Supplementary Materials

This PDF file includes:

Supplementary Text
Figs. S1 to S6
Legends for movies S1 to S4
Legends for data files S1 to S5

Other Supplementary Material for this manuscript includes the following:

Movies S1 to S4
Data files S1 to S5

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